

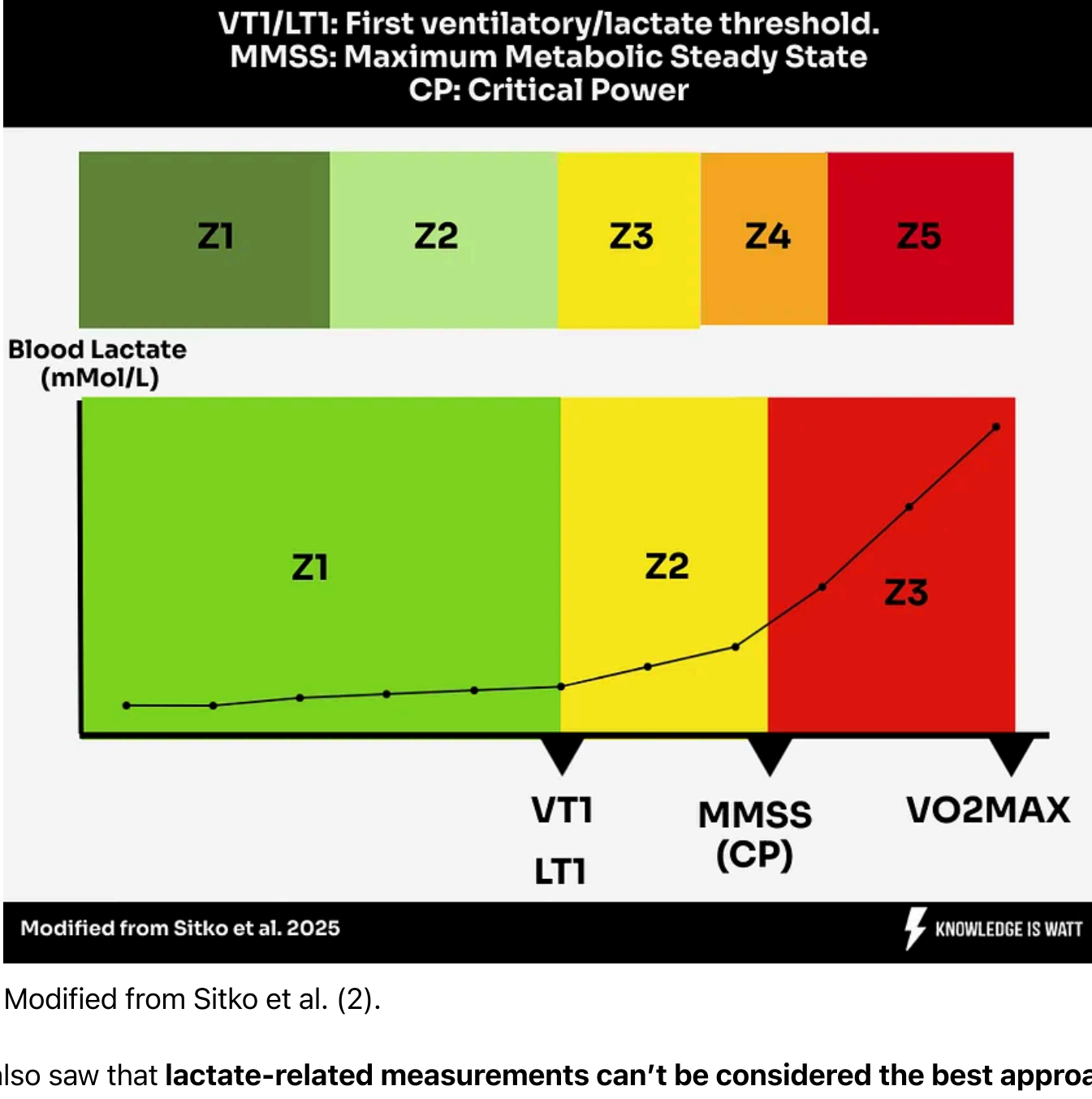
142. DOES FTP TRULY REPRESENT THE SECOND EXERCISE THRESHOLD?



In the last *KIW Research Notes*, we saw that, in 2026, **Critical Power (CP)** is the **gold standard for identifying the second exercise threshold**, also known as **Maximum Metabolic Steady State (MMSS)**, which is “*the bioenergetic threshold above which fatigue development is accelerated and both systemic and muscular homeostasis can no longer be maintained.*” (1)

Indeed, scientific studies have consistently shown that exercising just below or above Critical Power ($\pm 5/10\%$) elicits fundamentally different metabolic responses, corresponding respectively to the medium/heavy and high/severe exercise intensity domains.

- Below CP: oxygen uptake, lactate concentration, and ionic disturbance stabilize, allowing exercise to be sustained for a relatively long duration.
- On the other hand, above CP these variables can’t reach a steady-state and increase progressively until exhaustion.



Modified from Sitko et al. (2).

We also saw that **lactate-related measurements can’t be considered the best approach for estimating the 2nd exercise threshold (MMSS)**. Specifically:

- The Maximum Lactate Steady State (note the ‘L’ instead of ‘M’). which is the highest power at which the increase in blood lactate is less than 1 mmol/L between 10th and 30th minute of constant-load exercise, is a good estimate of the MMSS but is highly impractical. Why? Because it requires 3–5 days of testing, repeated prolonged efforts at near-threshold intensity, and a large number of lactate measurements.
- The Second Lactate Threshold determined from an incremental step test (LT2) cannot currently be considered an accurate estimate of MMSS. This is mainly because there is no combination of standardized reference protocol and LT2 definition which has consistently been shown to reliably estimate the MMSS.

Alternatively, **one of the most widely used methods to estimate the second threshold in cycling is the Functional Threshold Power**, more commonly known as **FTP**. This concept was introduced by **Andrew Coggan** in the early 2000s and is defined as **the highest mean power that an athlete can sustain for approximately one hour**. In practice, FTP is often estimated using a **20-minute time trial**, with FTP calculated as **95% of the average power achieved during the effort**. (3)

$$\text{FTP} = 20\text{min max power} - 5\%$$

The advantage of FTP testing is that it is very simple and practical to perform in the field:

- no blood lactate sampling
- only one maximal effort (20min), unlike CP determination, which typically requires 2/3 trials (3, 8 and 12 min for example).

FTP is the default metric used by many popular training platforms, such as Training Peaks and Zwift, to determine training intensity zones.

But an important question is:

Is FTP truly an accurate estimate of the second exercise threshold (MMSS)/Critical Power?

Enjoy the read!

As highlighted in a 2021 review of the scientific literature entitled “*What is known about the FTP20 test related to cycling?*” (4), only two studies have directly compared Functional Threshold Power (FTP) with Critical Power (CP)—currently considered the gold standard for identifying the 2nd exercise threshold (MMSS)—within the same group of athletes.

These two studies are:

- Morgan et al. (from University of Exeter, UK) published in 2019 on *Journal of Sport Sciences*. (5)
- Karsten et al. (from European University of Applied Sciences, Berlin) published in 2021 on *Frontiers in Physiology*. (6)

In both studies, the participants were trained amateur cyclists. Specifically:

- Morgan’s study: 12 club-level cyclists, ~25 years old, FTP ~275 W.
- Karsten’s study: 17 trained cyclists and triathletes ~30 years old, FTP ~250 W.

Both studies followed similar procedures, comparing within the same athletes:

- Critical Power (CP)** calculated from 3-5 high/severe intensity all-out trials
- FTP** calculated as 95% of the mean power achieved during a 20-minute all-out effort

Critical Power was calculated using the ‘best individual-fit’ approach, where the three common two-parameter CP models are tested and the model with the lowest estimation error is selected. This scientifically robust method can be automatically applied to your data using our *KIW CP/W’* calculator [here](#).

- When putting together all the data and calculating the average CP and FTP, the values were quite similar:**
 - Morgan’s study: 3 W or 2% average difference
 - FTP: 278 ± 42 W
 - CP: 275 ± 40 W
 - Karsten’s study: 7 W or 3% average difference
 - FTP: 249 ± 44 W
 - CP: 256 ± 50 W
- However, here is the main point: in both studies, at the individual level the differences between FTP and CP were large:**
 - ~50% of athletes showed differences >4%
 - Discrepancies arrived up to 15%

4 and 15% differences would mean:

→ For an athlete with CP = 275 W: 11 and 40 Watts

→ For an athlete with CP = 400 W: 15 and 60 Watts

- At the individual level, CP and FTP differ by approximately 4% on average, with discrepancies reaching up to 15% in some individuals.**

Scientific literature data seems very clear and coherent:

- At the individual athlete level, FTP doesn’t represent an accurate estimation of CP**, which in 2026 represents the gold standard for the 2nd exercise threshold (Maximum Metabolic Steady State), which distinguish between steady-state and non-steady state oxidative metabolism.
- So, **FTP and CP should not be used interchangeably**. Given the stronger physiological background, **Critical Power should be preferred over FTP when determining training zones, training load metrics and pacing strategies**.

Thanks for Reading! Share with your Team!

Gabriele Gallo, PhD in Exercise and Sport Sciences

Founder of Knowledgeiswatt

- Black MI, Jones AM, Blackwell JR, Bailey SJ, Wylie LJ, McDonagh ST, Thompson C, Kelly J, Sumners P, Mileva KN, Bowtell JL, Vanhatalo A. Muscle metabolic and neuromuscular determinants of fatigue during cycling in different exercise intensity domains. *J Appl Physiol* (1985). 2017 Mar 1;122(3):446-459.
- Sitko S, Artetxe X, Bonnevie-Svensden M, Galán-Rioja MÁ, Gallo G, Grappe F, Leo P, Mateo M, Mujika I, Sanders D, Seiler S, Zabala M, Valenzuela PL, Viribay A. What Is “Zone 2 Training”? Experts’ Viewpoint on Definition, Training Methods, and Expected Adaptations. *Int J Sports Physiol Perform*. 2025 Feb 26;20(11):1614-1617. doi: 10.1123/ijspp.2024-0303. PMID: 40010355.
- Allen H, Coggan A. Training and Racing With a Power Meter. Boulder, CO: Velopress; 2006
- Mackey J, Horner K. What is known about the FTP²⁰ test related to cycling? A scoping review. *J Sports Sci*. 2021 Dec;39(23):2735-2745. doi: 10.1080/02640414.2021.1955515. Epub 2021 Jul 26. PMID: 34304689
- Morgan PT, Black MI, Bailey SJ, Jones AM, Vanhatalo A. Road cycle TT performance: Relationship to the power-duration model and association with FTP. *J Sports Sci*. 2019 Apr;37(8):902-910. doi: 10.1080/02640414.2018.1535772. Epub 2018 Nov 2. PMID: 30387374.
- Karsten B, Petrigna L, Klose A, Bianco A, Townsend N and Triska C (2021) Relationship Between the Critical Power Test and a 20-min Functional Threshold Power Test in Cycling. *Front. Physiol*. 11:613151. doi: 10.3389/fphys.2020.613151